



University of Pittsburgh

Science and Commonsense

An Inferentialist Approach towards Primitivism of
Colors

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The Scientific Image and the Manifest Image

Eddington's two tables

Background

Introduction

- Overview
- Metaphysics of Colors

Irreducibility Arguments

- Putnam
- Benacerraf
- Gert on Colors
- My Reservation

Inferentialist Approach

- Overview
- Inferentialism
- Main Claim
- Fine Print

Conclusion

The Scientific Image and the Manifest Image

Eddington's two tables

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion



The Manifest Image

The Scientific Image and the Manifest Image

Eddington's two tables

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

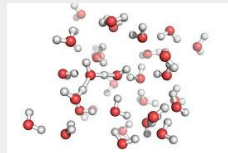
Main Claim

Fine Print

Conclusion



The Manifest Image



The Scientific Image

The Scientific Image and the Manifest Image

Eddington's two tables

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

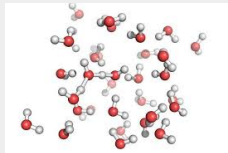
Main Claim

Fine Print

Conclusion



The Manifest Image



The Scientific Image

- What is the relation between these two images? That is, what is the relation between science and commonsense?

The Scientific Image and the Manifest Image

Eddington's two tables

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

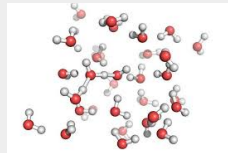
Main Claim

Fine Print

Conclusion



The Manifest Image



The Scientific Image

- What is the relation between these two images? That is, what is the relation between science and commonsense?



- How about the godly image?

The Scientific Image and the Manifest Image

Science versus Commonsense

Background

Introduction

- Overview
- Metaphysics of Colors

Irreducibility Arguments

- Putnam
- Benacerraf
- Gert on Colors
- My Reservation

Inferentialist Approach

- Overview
- Inferentialism
- Main Claim
- Fine Print

Conclusion

The Scientific Image and the Manifest Image

Science versus Commonsense

Background

Introduction

Overview
Metaphysics of
Colors

Irreducibility Arguments

Putnam
Benacerraf
Gert on Colors
My Reservation

Inferentialist Approach

Overview
Inferentialism
Main Claim
Fine Print

Conclusion

- Sellars(1956): “in the dimension of describing and explaining the world, science is the measure of all things, of what is that it is, and of what is not that it is not”

The Scientific Image and the Manifest Image

Science versus Commonsense

Background

Introduction

Overview
Metaphysics of
Colors

Irreducibility Arguments

Putnam
Benacerraf
Gert on Colors
My Reservation

Inferentialist Approach

Overview
Inferentialism
Main Claim
Fine Print

Conclusion

- ▶ Sellars(1956): “in the dimension of describing and explaining the world, science is the measure of all things, of what is that it is, and of what is not that it is not”
- ▶ McDowell: Science and commonsense are two separated domains that have very little to do with each other.

The Scientific Image and the Manifest Image

Science versus Commonsense

Background

Introduction

Overview
Metaphysics of
Colors

Irreducibility Arguments

Putnam
Benacerraf
Gert on Colors
My Reservation

Inferentialist Approach

Overview
Inferentialism
Main Claim
Fine Print

Conclusion

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 - ▶ *Hard to find an explicit assertion, but one can tell this attitude from, e.g. “Knowledge and the Internal” or “Mind and World”. Or personal communication.*

The Scientific Image and the Manifest Image

Science versus Commonsense

Background

Introduction

Overview
Metaphysics of
Colors

Irreducibility Arguments

Putnam
Benacerraf
Gert on Colors
My Reservation

Inferentialist Approach

Overview
Inferentialism
Main Claim
Fine Print

Conclusion

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- ▶ Quine(1957): “Science is not a substitute for common sense but an extension of it.”

The Scientific Image and the Manifest Image

Science versus Commonsense

Background

Introduction

Overview
Metaphysics of
Colors

Irreducibility Arguments

Putnam
Benacerraf
Gert on Colors
My Reservation

Inferentialist Approach

Overview
Inferentialism
Main Claim
Fine Print

Conclusion

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- ▶ Quine(1957): “Science is not a substitute for common sense but an extension of it.”
 - ▶ Spoiler Alert: I’m with Quine.

Color as an Entry-Point

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

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Color as an Entry-Point

- ▶ Overall, I'd like to discuss different views about science and commonsense in the context of colors.
 - ▶ Sufficient Scientific Understanding of Color Perception
 - ▶ Wide-spread in the manifest image but barely exist in the scientific image
 - ▶ Popular target for explaining-away projects (reduction, elimination, naturalization). E.g. Frank Jackson (1998, Ch 4), David Lewis (1997), Hardin (1993), etc.

Overview

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Overview

- ▶ Joshua Gert (2017) proposed a new argument for the irreducibility of colors, which, in the broader context, can be seen as a refusal of scientific realism/naturalism.

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- ▶ Me: Gert's argument forces him to adopt a McDowell-like attitude toward the relation between science and commonsense, making science and commonsense two separated domains that have little to do with each other.

Overview

- ▶ Joshua Gert (2017) proposed a new argument for the irreducibility of colors, which, in the broader context, can be seen as a refusal of scientific realism/naturalism.
- ▶ Me: Gert's argument forces him to adopt a McDowell-like attitude toward the relation between science and commonsense, making science and commonsense two separated domains that have little to do with each other.
- ▶ I propose an alternative argument for the irreducibility of colors, which exemplifies Quine's view, that science is an extension instead of a replacement of commonsense.

Oversimplified Introduction to Color Metaphysics Reductionism

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Oversimplified Introduction to Color Metaphysics Reductionism

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

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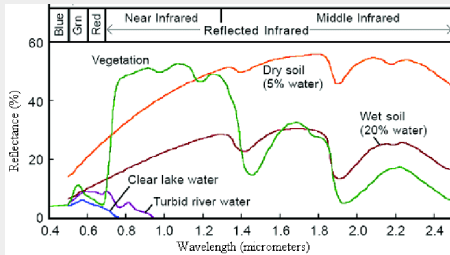
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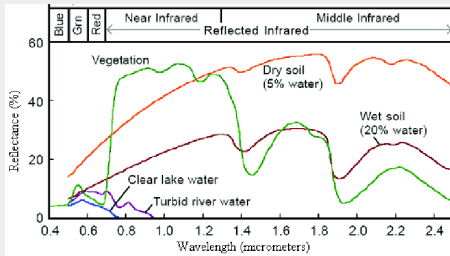
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- ▶ Problem: SSR is sometimes highly disjunctive, e.g. brown

Oversimplified Introduction to Color Metaphysics Eliminativism

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Oversimplified Introduction to Color Metaphysics

Eliminativism

- Some Prominent Supporters: Hardin (1988), Boghossian & Velleman (1989), Averill (2005), and Maund (1995, 2006, 2011)

Oversimplified Introduction to Color Metaphysics

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- ▶ Some Prominent Supporters: Hardin (1988), Boghossian & Velleman (1989), Averill (2005), and Maund (1995, 2006, 2011)
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Oversimplified Introduction to Color Metaphysics Primitivism

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

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Oversimplified Introduction to Color Metaphysics

Primitivism

- ▶ Some Prominent Supporters: Hacker(1987), J. Campbell (1994, 2005), McGinn (1996), Watkins (2005), etc.
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 - ▶ “Colors are irreducible, mind independent, *sui generis* properties of objects, and that any normal eight-year-old in the developed world knows which objects in their environment have which ones.” (Gert, 2017)

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- ▶ What’s new with Gert’s account?

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 - ▶ “Colors are irreducible, mind independent, *sui generis* properties of objects, and that any normal eight-year-old in the developed world knows which objects in their environment have which ones.” (Gert, 2017)
- ▶ What’s new with Gert’s account?
 - ▶ Gert provided a novel argument in defense of the irreducibility of certain notions in the manifest image

Ain't we here already?

Putnam (1975)

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

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Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

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Putnam (1975)

the explanatory autonomy of manifest concepts

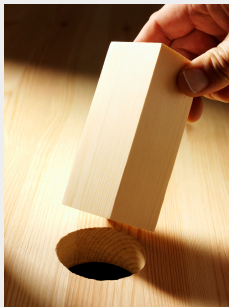
Shapes cannot be reduced to micro-physical properties since shapes are essential to some good explanations that one cannot give in terms of micro-physical properties.

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Ain't we here already? No. Fodor (1974)

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

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Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Separating Explanation from Ontology

Fodor would accept that color-types and shape-types do play essential roles in explanations that physical-types cannot play. But he can still hold an ontological token-reductionism of colors and shapes, asserting that every color-token or shape-token is identical to a physical-token.

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Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

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- Gert proposes that we should seek an argument for the irreducibility of colors that does not rely on the explanation issues.

Irreducibility of Numbers

Benacerraf (1973)

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Irreducibility of Numbers

Benacerraf (1973)

Number	Reduction-1	Reduction-2
0	$\emptyset$	$\emptyset$
1	$\{\emptyset\}$, i.e. $\{0\}$	$\{\emptyset\}$, i.e. $\{0\}$
2	$\{\emptyset, \{\emptyset\}\}$, i.e. $\{0, 1\}$	$\{\{\emptyset\}\}$, i.e. $\{1\}$
3	$\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$, i.e. $\{0, 1, 2\}$	$\{\{\{\emptyset\}\}\}$, i.e. $\{2\}$

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Two Reductions Disagree

Reduction-1 attributes to numbers the following claim: $1 \in 3$.

Irreducibility of Numbers

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Reduction-1 attributes to numbers the following claim: $1 \in 3$.

Reduction-2 attributes to numbers the following claim: $1 \notin 3$.

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Two Reductions Disagree

Reduction-1 attributes to numbers the following claim: $1 \in 3$.

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Which is true?

Irreducibility of Numbers

Benacerraf (1973)

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Irreducibility of Numbers

Benacerraf (1973)

Neither reduction is valid.

The problem Benacerraf finds of reductions of numbers is that either reduction would attribute to numbers additional properties that do not apply to numbers.

Irreducibility of Numbers

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- Gert wants to say the same about colors. Namely, any reduction of color would attribute to colors properties or claims that do not hold of colors.

Gert: The Irreducibility of Colors

Step 1: Modal Properties

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Gert: The Irreducibility of Colors

Step 1: Modal Properties

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Gert: The Irreducibility of Colors

Step 1: Modal Properties

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Problematic Attribution: Too many modal properties

According to Gert (2008), identifying a color, say red, with certain physical property, say a certain set of SSR, will attribute many modal claims to red.

Gert: The Irreducibility of Colors

Step 1: Modal Properties

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

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Gert: The Irreducibility of Colors

Step 1: Modal Properties

Problematic Attribution: Too many modal properties

According to Gert (2008), identifying a color, say red, with certain physical property, say a certain set of SSR, will attribute many modal claims to red.

- ▶ I didn't quite find a concrete example of such alleged modal properties/claims.
- ▶ But the rough idea should be clear. Suppose we have a very sophisticated science of SSRs. With the help of that science, we would be able to find many modal properties/claims of SSRs. Then we can translate those properties/claims to colors.

Gert: The Irreducibility of Colors

Discovery of Hidden Nature?

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Gert: The Irreducibility of Colors

Discovery of Hidden Nature?

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

- ▶ A determined reductionist can grant that any reduction of colors will attribute additional properties/claims to colors and insist that such identification should be **celebrated** for discovering unknown **hidden natures** of numbers.

Gert: The Irreducibility of Colors

Discovery of Hidden Nature?

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

- ▶ A determined reductionist can grant that any reduction of colors will attribute additional properties/claims to colors and insist that such identification should be **celebrated** for discovering unknown **hidden natures** of numbers.
- ▶ Gert responds to this concern by appealing to Neo-Pragmatism.

Gert: The Irreducibility of Colors

Step 2: Neo-Pragmatism

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Gert: The Irreducibility of Colors

Step 2: Neo-Pragmatism

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Neo-pragmatism

Neo-pragmatism is a recent program of philosophy, advocated by Huw Price among others, emphasizing linguistic practices. The relevant bit of the program here is that it denies substantial notions of truth and reference and proposes deflationary alternatives . For neo-pragmatists, truth and reference are not things out there to be grasped by us. Rather they are grounded in our usage of language

Gert: The Irreducibility of Colors

Step 2: Neo-Pragmatism

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

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- Gert: Those attributed modal claims are not grounded in our linguistic practices, and thus are not true. There is no hidden nature of colors to be discovered by science.

Main Reservation

The Separation of Science and Commonsense

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility

Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist

Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Main Reservation

The Separation of Science and Commonsense

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

- For Gert, our daily usage of color terms contains all that there is to be known about colors. Then, what have color scientists been doing?

Main Reservation

The Separation of Science and Commonsense

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

- ▶ For Gert, our daily usage of color terms contains all that there is to be known about colors. Then, what have color scientists been doing?
- ▶ Gert's view, taken literally, suggests that the concepts of colors as used in ordinary language not only barely have anything to do with the concepts of colors as used in color sciences, despite color scientists also call what they study as colors.

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The Separation of Science and Commonsense

- ▶ For Gert, our daily usage of color terms contains all that there is to be known about colors. Then, what have color scientists been doing?
- ▶ Gert's view, taken literally, suggests that the concepts of colors as used in ordinary language not only barely have anything to do with the concepts of colors as used in color sciences, despite color scientists also call what they study as colors.
- ▶ This ignores the fact that scientists are also ordinary people who are part of our linguistic community. They do use the same concepts as we do even in conduct of color science.

The Idea in a Nutshell

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

The Idea in a Nutshell

An Observation/Assertion

Given that I'm a person who has a normal color perception capacity and knows how to use color concepts as most twelve-year-old do, should I perceive an ordinary banana under a normal lighting condition to be yellow, no scientist can convince me otherwise. (And it is rational of me to do so.)

The Idea in a Nutshell

An Observation/Assertion

Given that I'm a person who has a normal color perception capacity and knows how to use color concepts as most twelve-year-old do, should I perceive an ordinary banana under a normal lighting condition to be yellow, no scientist can convince me otherwise. (And it is rational of me to do so.)

The Irreducibility of Colors Follows.

If colors are, in fact, surface spectral reflectances, or some other physical, perhaps highly disjunctive, properties, science would be the final tribunal for their assessments. Thus, rationality would require me to defer to color scientists' judgment, should we disagree, in the case above.

Preliminaries on Inferentialism

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Preliminaries on Inferentialism

Inferential Relations as Primitive

Propositions/Judgments are connected to each other under semantic relations: reason-for and reason-against.

- ▶ Other notions, e.g. truth, are to be derived from inferential relations.

Preliminaries on Inferentialism

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Reason For $\vdash$

“I see this table as red” $\vdash$ “This table is red”

Preliminaries on Inferentialism

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Reason For $\vdash$

“I see this table as red” $\vdash$ “This table is red”

Reason Against $\#$

“I see this table as green” $\#$ “This table is red”

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Inferential Relations as Primitive

Propositions/Judgments are connected to each other under semantic relations: reason-for and reason-against.

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Reason For $\vdash$

“I see this table as red” $\vdash$ “This table is red”

Reason Against $\#$

“I see this table as green” $\#$ “This table is red”

These Relations can be defeated, i.e. not monotone

“I see this table as red”, “I’m color-blind” $\nvdash$ “This table is red”

Main Claim

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Main Claim

Autonomy of Color Judgments (in words)

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility
Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist
Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Main Claim

Autonomy of Color Judgments (in words)

- For any given color judgment, all propositions are that **directly** connected to that judgment via the reason for/against relation, are propositions in (the vocabulary of) the manifest image.

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- ▶ For any given color judgment, all propositions are that **directly** connected to that judgment via the reason for/against relation, are propositions in (the vocabulary of) the manifest image.
- ▶ E.g. the agent's perceptual experience, the lighting condition, the agent's perceptual capacities, the context of color perception, among other relevant commonsensical considerations that any reflective person without much knowledge of physics or vision science can have.

Main Claim

Autonomy of Color Judgments (in words)

- ▶ For any given color judgment, all propositions are that **directly** connected to that judgment via the reason for/against relation, are propositions in (the vocabulary of) the manifest image.
- ▶ E.g. the agent's perceptual experience, the lighting condition, the agent's perceptual capacities, the context of color perception, among other relevant commonsensical considerations that any reflective person without much knowledge of physics or vision science can have.
- ▶ These commonsensical considerations give a decisive verdict on the truth/falsity of the given color judgment under consideration, in the sense that no further claim of science can defeat such a reason for/against relation.

Main Claim

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

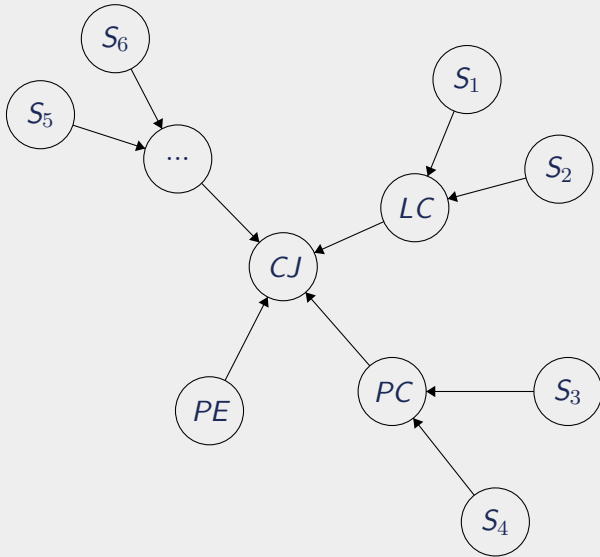
Inferentialism

Main Claim

Fine Print

Conclusion

Main Claim



Fine Print

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Fine Print

Def. (Reason For)

Proposition A is a *reason for* C iff there is some Γ such that $\Gamma \cup A \vdash C$ and $\Gamma - A \not\vdash C$.

Fine Print

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Def. (Indirect Reason)

Proposition A is an *indirect* reason for C iff (1) A is a reason for C ; (2) there is a proposition B that *screen-off* the influence of A on C .

Fine Print

Def. (Reason For)

Proposition A is a *reason for* C iff there is some Γ such that $\Gamma \cup A \vdash C$ and $\Gamma - A \not\vdash C$.

Def. (Indirect Reason)

Proposition A is an *indirect* reason for C iff (1) A is a reason for C; (2) there is a proposition B that *screen-off* the influence of A on C.

Def. (Direct Reason)

Proposition B is a direct reason for proposition C iff (1) B is a reason for C and (2) B is not an indirect reason for C.

Fine Print Cont.

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Fine Print Cont.

Def. (Screen off)

Proposition B screen-off the influence of A on C iff for any Γ that contains B, (1) $\Gamma \vdash C \implies (\Gamma \cup \{A\} \vdash C \text{ and } \Gamma \cup \{A\} \vdash C)$ and (2) $\Gamma \not\vdash C \implies (\Gamma \cup \{A\} \not\vdash C \text{ and } \Gamma \cup \{A\} \not\vdash C)$.

Fine Print Cont.

Def. (Screen off)

Proposition B screen-off the influence of A on C iff for any Γ that contains B, (1) $\Gamma \vdash C \implies (\Gamma \cup \{A\} \vdash C \text{ and } \Gamma \cup \{A\} \vdash C)$ and (2) $\Gamma \not\vdash C \implies (\Gamma \cup \{A\} \not\vdash C \text{ and } \Gamma \cup \{A\} \not\vdash C)$.

- Intuitively, proposition B screen-off the influence of A over C if and only if for every set Γ containing B, adding or removing A to or from Γ doesn't change the reason for/against relation holding between Γ and C. That is, in the presence of B, the influence of A on C is silenced.

Main Claim

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Main Claim

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Main Claim

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Autonomy of Color Judgments (in technical words)

- ▶ For any given color judgment C , all propositions are direct reasons for/against C propositions in (the vocabulary of) the manifest image.
- ▶ The set of all direct reasons of C gives a decisive verdict on the truth/falsity of C , beyond the revision of scientific discoveries.

Main Claim

Autonomy of Color Judgments (in technical words)

- ▶ For any given color judgment C , all propositions are direct reasons for/against C propositions in (the vocabulary of) the manifest image.
- ▶ The set of all direct reasons of C gives a decisive verdict on the truth/falsity of C , beyond the revision of scientific discoveries.
- ▶ Propositions in the scientific image do provide reasons for/against color judgments, but only via those direct manifest reasons.

Implication

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Implication

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

No Hidden Nature

Implication

No Hidden Nature

- ▶ My account makes explicit the precise sense in which there is no hidden nature of color to be discovered by science.

Implication

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- ▶ My account makes explicit the precise sense in which there is no hidden nature of color to be discovered by science.
- ▶ The truth/falsity of a given color judgment is ultimately determined by a certain set of manifest considerations.

Implication

No Hidden Nature

- ▶ My account makes explicit the precise sense in which there is no hidden nature of color to be discovered by science.
- ▶ The truth/falsity of a given color judgment is ultimately determined by a certain set of manifest considerations.
- ▶ In this sense, there is no hidden nature of color to be discovered by science, despite that scientific discoveries can and do from time to time revise my judgments of colors (through those manifest considerations).

Implication

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Irreducibility via Modus Tollens

Implication

Background

Introduction

Overview

Metaphysics of
Colors

Irreducibility Arguments

Putnam

Benacerraf

Gert on Colors

My Reservation

Inferentialist Approach

Overview

Inferentialism

Main Claim

Fine Print

Conclusion

Irreducibility via Modus Tollens

My account refutes the identification of colors with any physical properties. If colors are, in fact, certain physical, albeit highly disjunctive, properties, science would be the final tribunal for their assessments.

Implication

Irreducibility via Modus Tollens

My account refutes the identification of colors with any physical properties. If colors are, in fact, certain physical, albeit highly disjunctive, properties, science would be the final tribunal for their assessments.

Science and Commonsense

My account doesn't assert that scientific claims/discoveries cannot be reasons for/against certain color judgments. Indeed, according to my account, they can. But they do so only via commonsensical considerations.

Thanks!

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